

STM32F446RE Based Motor Health Monitoring System

1. Project Overview

The Motor Health Monitoring System continuously monitors motor current, temperature, and vibration to detect abnormal conditions and generate fault alerts. The STM32F446RE performs sensor acquisition, fault analysis, and communication with an ESP32 module for future IoT integration.

2. STM32F446RE Overview

STM32F446RE is a 32-bit ARM Cortex-M4 microcontroller operating up to 180 MHz with 512 KB Flash, 128 KB SRAM, multiple ADC channels, UART interfaces, timers, and GPIOs suitable for industrial monitoring applications.

3. Sensor Acquisition Using ADC

Three analog sensors are connected to ADC channels. Current Sensor -> ADC1_IN0, Temperature Sensor -> ADC1_IN1, and Vibration Sensor -> ADC1_IN2. Sensor values are periodically sampled and converted into engineering units for analysis.

4. UART Communication with ESP32

UART communication is used to transfer motor health data and fault notifications from STM32 to ESP32. Communication settings: 115200 baud, 8 data bits, no parity, 1 stop bit.

5. Fault Detection Logic

GOOD: Temperature < 70°C, Current < 15A, Vibration < 60%. WARNING: Temperature 70–90°C, Current 15–18A, Vibration 60–80%. FAULT: Temperature > 90°C, Current > 18A, or Vibration > 80%. Fault conditions activate LEDs, buzzer, and UART alerts.

6. GPIO Mapping

PA0 - Current Sensor Input PA1 - Temperature Sensor Input PA2 - Vibration Sensor Input PA9 - UART TX PA10 - UART RX PB0 - Motor LED PB1 - Warning LED PB2 - Fault LED PB10 - Buzzer Output

7. System Flow

Sensors -> STM32 ADC Acquisition -> Fault Detection Algorithm -> GPIO Outputs (LEDs/Buzzer) -> UART Communication -> ESP32 -> Future Cloud Dashboard

8. Conclusion

The STM32F446RE-based Motor Health Monitoring System provides real-time monitoring, fault detection, alert generation, and a foundation for predictive maintenance and IoT integration.